

ReactJS HOL-10

Objective:

Develop a React application named **officespacereentalapp** using **JSX** to display office space details, render multiple office objects, and apply inline CSS based on the rent amount.

Theory

1. Define JSX

JSX (JavaScript XML) is a syntax extension that allows HTML-like code to be written inside JavaScript. React converts JSX into `React.createElement()` calls.

2. Explain ECMAScript (ES6)

ECMAScript 6 introduced modern JavaScript features such as `let`, `const`, arrow functions, classes, modules, template literals, destructuring, and the spread operator.

3. Explain `React.createElement()`

`React.createElement()` creates React elements that represent the UI. JSX is internally compiled into `React.createElement()` calls.

4. Creating React Nodes with JSX

React nodes are created by writing JSX elements such as `<h1>`, `<div>`, and custom React components.

5. Rendering JSX to the DOM

ReactDOM renders JSX into the browser using `ReactDOM.createRoot(...).render(<App />)`.

6. JavaScript Expressions in JSX

Expressions are written inside curly braces `{}` and evaluated dynamically.

7. Inline CSS in JSX

Inline styles are JavaScript objects passed to the `style` attribute, e.g., `style={{color:'red'}}`.

Prerequisites

- Node.js
- npm
- Visual Studio Code

Implementation Steps

1. Create a React project named officespacerentalapp.
2. Create an array of office objects with Name, Rent, Address and Image.
3. Display the heading using JSX.
4. Display office images and details.
5. Loop through office objects using map().
6. Apply inline CSS to show rent in red if below ₹60,000, otherwise green.
7. Run the application using npm start.

App.js

```
import officeImg from './office.jpg';

function App() {
  const offices=[
    {Name: 'DBS',Rent:50000,Address:'Chennai',Image:officeImg},
    {Name: 'TCS',Rent:75000,Address:'Hyderabad',Image:officeImg}
  ];

  return (
    <div>
      <h1>Office Space, at Affordable Range</h1>
      {offices.map((item,index)=>(
        <div key={index}>
          <img src={item.Image} alt='Office' width='250' />
          <h2>Name: {item.Name}</h2>
          <h3 style={{color:item.Rent<60000?'red':'green'}}>
            Rent: Rs. {item.Rent}
          </h3>
          <h3>Address: {item.Address}</h3>
        </div>
      ))}
    </div>
  );
}
export default App;
```

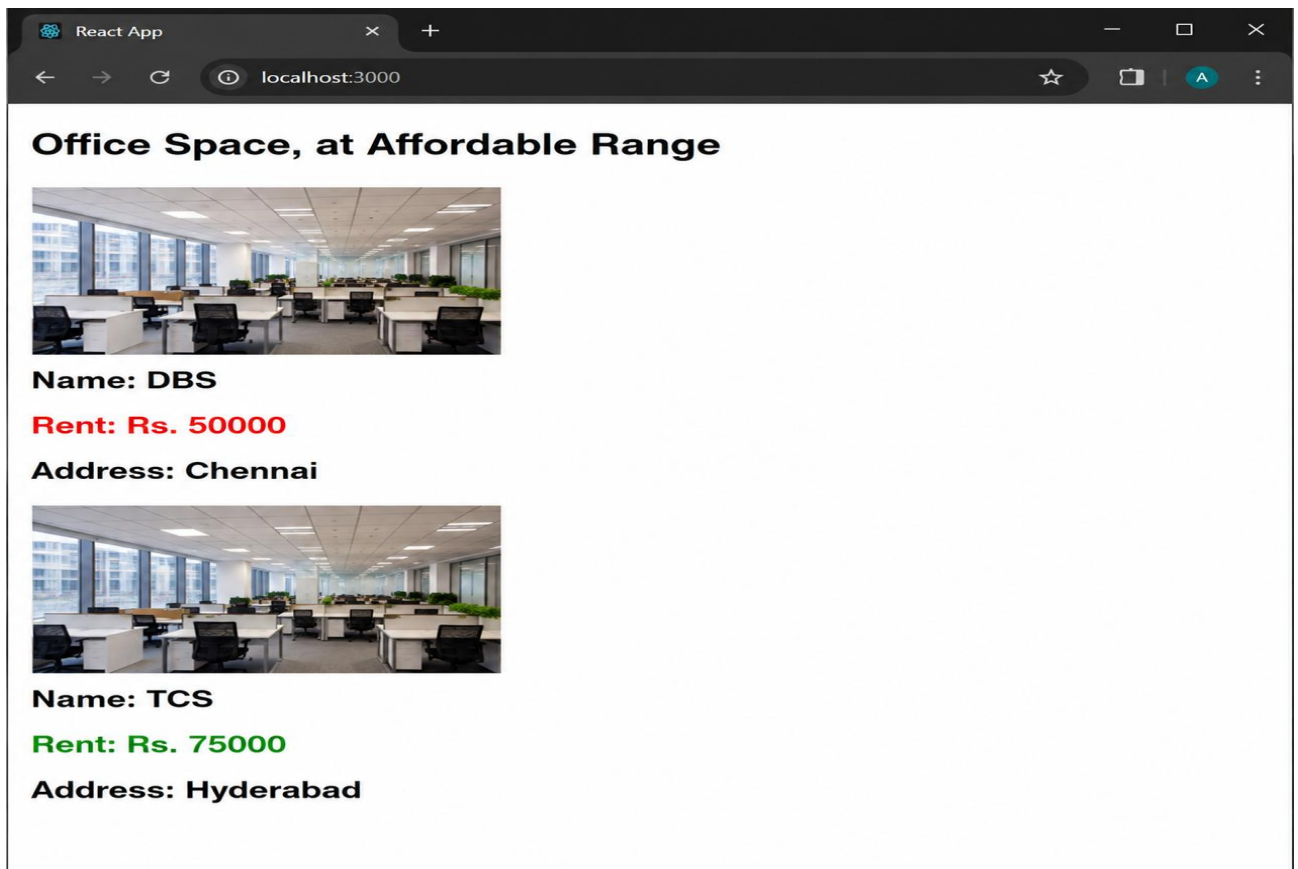
index.js

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';

const
root=ReactDOM.createRoot(document.getElementById('root'));
root.render(<App />);
```

Output Screenshot:

- Displays the heading 'Office Space, at Affordable Range'.
- Shows office images.
- Displays Name, Rent, and Address for each office.
- Rent is shown in red when below ₹60,000 and green when ₹60,000 or above.



Learning Outcome

Successfully used JSX, JavaScript expressions, inline CSS, arrays, `map()`, and conditional styling in a React application.

Conclusion

This exercise demonstrated how JSX simplifies UI creation and how React can dynamically render data with conditional formatting.